Prodigy Anesthesia Quick Review

Thoracic Surgery

Thoracic surgery commonly involves fiber optic bronchoscopy. What is the smallest endotracheal tube (ETT) in which a standard fiber optic bronchoscope will pass?	8.0 *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 278.*
What is the most appropriate form of treatment recommended for patients with early-stage lung cancer?	Surgery. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 281.*
What perioperative test is an adequate assessment of pulmonary function in patients presenting for thoracic surgery who have very little or absent functional impairment?	Spirometry *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282.*
Patients revealing what spirometric values warrant further testing prior to pulmonary resection?	Patients with the following spirometric values warrant further testing prior to surgery: FEV1 (forced expired volume in 1 second) < 60% predicted, FVC (forced vital capacity) < 60% predicted, Maximum voluntary ventilation < 50% predicted, FEV1/FVC ratio < 50%, Diffusing capacity for carbon monoxide < 60% predicted, Arterial carbon dioxide tension > 45mmHg. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282.*
The majority of patients undergoing pneumonectomy or lobectomy are placed in what surgical position?	Lateral decubitus position. This position allows a posterolateral or lateral thoracotomy, permitting the best exposure of the pulmonary hilum. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282.*
What hemodynamic changes can result when the pericardium or hilar structures are retracted vigorously?	Cardiac arrhythmias and hypotension can occur. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282.*

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Thoracic Surgery

The surgeon asks the anesthetist to reinflate the lung while he clamps the bronchus that leads to the lobe being resected. What is the purpose of this?	To ensure that the remaining lobes inflate adequately. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282.*
If a chest tube is placed following a pneumonectomy, why is a balanced drainage system imperative?	A mediastinal shift towards the operative side will occur if a balanced drainage system is not used, resulting in severe hemodynamic changes. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 282-283.*
List 3 relative contraindications to lung resection.	1. Right ventricular dysfunction 2. Pulmonary hypertension 3. Preoperative hypercapnia *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 285.*
What are the most common preoperative diagnoses seen in patients presenting for lobectomy or pneumonectomy?	The most common preoperative diagnosis is carcinoma of the lung, however patients may also present with infection (namely bronchiectasis and mycobacterial disease), developmental abnormalities, and trauma. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 283.*
A patient presenting for lung resection has myasthenic syndrome (Eaton-Lambert syndrome). How would you expect the patient to respond to depolarizing muscle relaxants? Nondepolarizing muscle relaxants?	The patient would exhibit resistance to depolarizing muscle relaxants and an increased sensitivity to nondepolarizing muscle relaxants. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 285.*
What are the benefits to the intraoperative use of regional analgesia during a lobectomy or pneumonectomy?	The amount of systemic analgesics and anesthetic is reduced, thus facilitating a quick emergence. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 286-287.*
Why is it important not to overhydrate a patient during a lobectomy or pneumonectomy?	Following surgery, the pulmonary vascular resistance increases relative to the portion of removed lung tissue. Overhydration places these patients at risk for pulmonary edema and right ventricular failure. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 287.*

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Thoracic Surgery

1. What are the most common sized double-lumen tubes (DLT) used in men? 2. What are the most common sized DTLs used in wormen?	1. 39-41 Fr 2. 35-37 Fr *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 287.*
What is a wedge resection? For what reasons would a wedge resection be indicated?	A wedge resection is the excision of a mass in a way that does not involve total removal of an entire pulmonary segment. Wedge resections may be perfomed in those with peripheral non-small-cell tumors, patients with extremely limited pulmonary reserve who could not tolerate a lobectomy, for the resection of multiple or single metastatic lesions arising from varying primary neoplasms, therapeutic and diagnostic purposes in lesions that cannot be diagnosed by less invasive techniques. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 290.*
What are the two most common indications for chest-wall resection.	1. Primary chest-wall tumors 2. Lung cancers that have invaded the thoracic cage *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 293.*
A patient presenting for chest-wall resection has received chemotherapy with Adriamycin preoperatively. What are the concerns associated with this chemotherapeutic agent?	At high doses, Adriamycin is associated with cardiotoxicity. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 293.*
What is empyema? How is it primarily treated?	Empyema is a collection of pus and fluid within the pleural cavity. Drainage of the infected fluid is the primary treatment. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 477.*
Empyema most commonly occurs secondary to what disorder?	Pneumonia *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 300.*
List the three phases of empyema.	1. Exudative phase 2. Fibrinopurulent phase 3. Organized phase. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 300.*

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Thoracic Surgery

How is empyema diagnosed?	Thoracentesis *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 300.*
What signs and symptoms are associated with the exudative phase of empyema?	Dyspnea, fever, and pleural effusion *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 300.*
What is the treatment for patients with extensive, established empyema?	Open thoracotomy. Blood and flud losses can be sizeable due to extensive intrapleural inflammation. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 301.*
What other underlying conditions are often associated with patients with empyema.	Sepsis, malnutrition, bronchopleural fistula *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 302.*
What are common preoperative pulmonary findings in patients presenting for drainage of empyema?	Mediastinal shift to the ipsilateral side, ipsilateral lung collapse, impaired hypoxic pulmonary vasoconstriction due to infection *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 302.*
Proper tube position of double-lumen tubes is always important. What is of particular concern in patients with empyema and proper tube placement?	If the double lumen tube is not in proper position, the nonaffected lung can become contaminated by the affected side. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 303.*
What are the two primary indications for tracheal resection?	1. Benign strictures (frequently from prior intubations or tracheostomy) 2. Primary tracheal neoplasms, most often adenoid cystic carcinoma and squamous cell carcinoma *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 304.*
Where would the incision be made for patients presenting for tracheal resection with lesions of the mid to upper trachea?	The neck *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 304.*

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Thoracic Surgery

Where would the incision be made for patients presenting for tracheal resection with lesions of the carina and lower trachea?	The right chest *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 304.*
How should a patient be placed after tracheal resection?	The patient should be placed head-up with th neck flexed. Flexion of the neck reduces tension placed on the tracheal suture line. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 308.*
Describe the typical patient population presenting for excision of apical blebs.	These patients typically are young, otherwise healthy individuals, with normal lung function. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 334.*
What conditions are frequently seen in patients with bullous emphysema?	Pulmonary hypertension, right ventricular dysfunction, end-stage lung disease. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 334.*
Are the majority of blebs or bullae excised through open procedures or thoracoscopically?	Thoracoscopically. The goal of the procedure (regardless of technique used) is to remove the nonfunctioning bleb or bullae. This allows the compressed, preserved lung tissue to reinflate, once again contributing to gas exchange. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 333-334.*
What is a major concern for patients undergoing excision of bilateral, large bullae?	Tension pneumothorax/pulmonary tamponade due to the expansion of bullae from positive pressure ventilation. It is best to isolate the affected lung before positive pressure ventilation is initiated. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 335.*
What measures can be taken to minimize the risk of bulla rupture during positive pressure ventilation?	Use pressure-controlled ventilation <20 cmH2O and low tidal volumes *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 335.*
Describe the typical patient presenting for lung-volume reduction surgery.	These patients have severe emphysema, require steroids, use supplemental oxygen and bronchodilators. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 336.*

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Thoracic Surgery

How does lung-volume reduction surgery improve pulmonary function?

By reducing lung volume through the removal of diseased tissue, elastic recoil improves and airway resistance is decreased. The chest cavity itself is reduced, therefore diaphragmatic and chest-wall function is improved.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 336.

What two surgical approaches can be used for lung-volume reduction surgery?

Median sternotomy, endoscopically

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 337.

Besides using an FiO2 of 100%, what is the most effective way to increase PaO2 level during one lung ventilation?

Applying CPAP to the nondependent lung.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1052.

Whether lung-volume reduction surgery is performed open or endoscopically, the patient should be extubated in the OR. Why is this?

To avoid unnecessary ventilatory pressures being placed on the lungs.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 337.

Following lung-volume reduction surgery, why are chest tubes left to water seal and not suction?

To prevent the build up of negative pressure on the lung that could disrupt staple lines.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 337.

Why would a patient undergo a bronchopulmonary lavage?

Bronchopulmonary lavage is performed in patients with pulmonary alveolar proteinosis. This condition is characterized by abnormal surfactant production that results in saturation of the lungs with proteinaceous fluid. The cause of this disorder is not well known.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 340.

Describe the process of a bronchopulmonary lavage.

Bronchopulmary lavage necessitates general anesthesia with the use of a double-lumen tube. The lung is irrigated with 500-1000 cc's of normal or half normal saline. This lavage washes out and dilutes the excess surfactant, mucus, or pus, and it also obtains specimens for histochemical and cytological examination.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 340.

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Thoracic Surgery

Pulmonary alveolar proteinosis is the the most common diagnosis seen in patients presenting for bronchopulmonary lavage. What are other possible preoperative diagnoses seen?	Cystic fibrosis, refractory asthma, inhalation of radioactive dust, silicosis, alveolar microlithiasis, lipoid pneumonitis, bronchiectasis *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 340.*
Typically, unilateral lavage is performed, with the more diseased lung irrigated first. If both lungs are equally diseased, what lung is irrigated first?	The left lung is irrigated first. The right lung is larger and can be used for ventilation to provide more adequate gas exchange. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 341.*
What signs and symptoms are seen in a person with pulmonary alveolar proteinosis?	Fatigue, dyspnea, cough, cyanosis, clubbing, diffuse rales, decreased pulmonary compliance *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 341.*
What would an ABG reveal in a person with pulmonary alveolar proteinosis?	Repiratory alkalosis and hypoxemia *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 341.*
List the most common indications for a lung transplant.	Pulmonary fibrosis, emphysema, pulmonary hypertension, cystic fibrosis. People with pulmonary hypertension typically have normal lung mechanics but extremely abnormal cardiac function. Patients with cystic fibrosis, pulmonary fibrosis, and emphysema have profoundly abnormal mechanical pulmonary function. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 344.*
Is cardio-pulmonary bybass (CBP) required for lung transplantation?	No, however a CPB circuit and perfusionist should be available at all times. *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 344.*
Upon reperfusion of the donor lung, should the anesthesia provider expect to see hypertension or hypotension?	Hypotension that usually spontaneously resolves *Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 344.*

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